TM Forum Component

Service Catalog Management

TMFC006

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Notice

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TMFC006 – Service Catalog Management

1. Overview

Component Name	ID	Description	ODA Function Block
Service Catalog Management	TMFC006	This Component has conformance testing for v4 APIs only at this point. The Service Catalog Management component is responsible for organizing the collection of service specifications that identify and define all requirements of a service that can be performed. This component has the functionality that enables presentation of a customer-facing view so users are able to search and select services they need, as well as a technical view to enable definition and setup of what is needed to deliver service specifications (Customer Facing Service Specifications (CFSSs) and Resource Facing Service Specifications (RFSSs)) contained in the service catalog. The Service Catalog Management component has functionalities that include creation of new service specifications, managing service specifications, administering the lifecycle of services, describing relationships between service specification attributes, reporting on service specifications and their changes, and facilitating easy and systematic indexing and access to services, as well as facilitating automation of the service delivery process.	Production

2. eTOM Processes, SID Data Entities and Functional Framework Functions

2.1. eTOM business activities

eTOM business activities this ODA Component is responsible for:

Identifier	Level	Business Activity Name	Description
1.4.13	L2	Service Catalog Lifecycle Management	Catalog Lifecycle Management business process covers a set of business activities that enable manage the lifecycle of an organizations catalog from design to build according to defined requirements.
1.4.14	L2	Service Catalog Operational Readiness Management	Service Catalog Operational Readiness Management business process establishes and administers the support needed to operationalize Service catalogs for ongoing day-to-day business needs.
1.4.15	L2	Service Catalog Content Management	Service Catalog Content Management business process define and provide the business activities that support the day-to-day operations of Service Catalogs in order to realize the business operations goals.
1.4.16	L2	Service Catalog Planning Management	Service Catalog Planning Management business process covers a set of business activities that understand and enable establish the plan to define, design and operationalize a catalog in order to meet the needs and objectives of Service cataloging. / The Service Catalog Planning Management business process ensure that the organization is able to identify the most appropriate scheme and goal for it catalog. It includes designing the Catalog plan and developing the specification according to Service management requirement.

1.4.19	L2	Service Specification Management	Service Specification Management business process leverages captured service requirements to develop, master, analyze, and update documented standard conditions that must be satisfied by service design and/or delivery. / Service Specifications Management can result in establishing, in a centralized way, technical (know-how) standards. / Such standards provide the organization with a means to control and approve the values and inputs of service specification through structure, review, approval and distribution processes to stakeholders and suppliers.
1.4.3	L2	Service Specification Development & Retirement	Service Specification Development & Retirement processes are project oriented in that they develop and deliver new or enhanced service types. These processes include process and procedure implementation, systems changes and customer documentation. They also undertake rollout and testing of the service type, capacity management and costing of the service type. It ensures the ability of the enterprise to deliver service types according to requirements.

1.4.3.4	L3	Develop Detailed Service Specifications	The Develop Detailed Service Specifications processes develop and document the detailed service-related technical and operational specifications, and customer manuals. These processes develop and document the required service features, the specific underpinning resource requirements and selections, the specific operational, and quality requirements and support activities, any service specific data required for the systems and network infrastructure as agreed through the Develop New Service Business Proposal processes. The Develop Detailed Product Specifications processes provide input to these specifications. The processes ensure that all detailed specifications are produced and appropriately documented. Additionally, the processes ensure that the documentation is captured in an appropriate enterprise repository.

2.2. SID ABEs

SID ABEs this ODA Component is responsible for:

SID ABE Level 1	SID ABE Level 2 (or set of BEs)	SID ABE L1 Definition	SID ABE L2 Definition

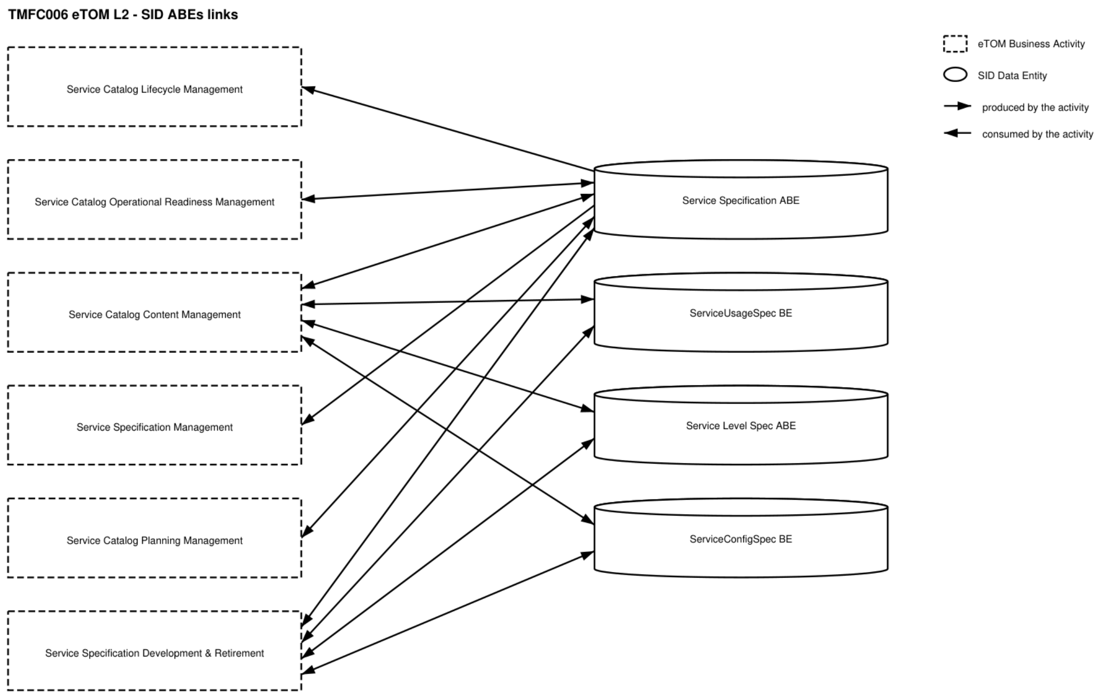
Service Specification		The Service Specification ABE contains entities that define the shared characteristics of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the characteristics defined in its associated specification. The Service Specification ABE is composed of two main concepts: The CSP Know-Hows specification a.k.a. Customer Facing Services Specification or CFSSpec that describe only functional characteristics and capabilities that the CSP can provide to a customer however the CSP builds on its own all or part of the technical solution. Note that it corresponds only to non-tangible goods. The Technical solution specification a.k.a. Resource Facing Services Specification or RFSpec that describes only technical characteristics and capabilities. Entities in this ABE focus on adherence to standards, distinguishing features of a Service, dependencies (both physical and logical, as well as on other services), quality, and cost. In general, entities in this ABE enable Services to be bound to Products and run using Resources.	*(no description available)*

Service Performance	Service Level Spec	The Service Performance ABE collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Know-Hows (a.k.a. CFS) and Technical Solutions (a.k.a. RFS). It provides a set of entities that enables performance monitoring and reporting. Each of these entities also enables network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also enable the traffic trend analysis.	The Service Specification ABE contains entities that define the invariant characteristics and behavior of both types of Service entities. This enables multiple instances to be derived from a single specification entity. In this derivation, each instance will use the invariant characteristics and behavior defined in its associated template. This ABE includes a third type of Service Specification entity: that of a ServiceLevelSpecification. This type of specification templatizes Services that are bound to a Service Level Agreement.
Service Performance	Service Performance Specification	The Service Performance ABE collects information used to correlate, consolidate, and validate various performance statistics and other operational characteristics of Know-Hows (a.k.a. CFS) and Technical Solutions (a.k.a. RFS). It provides a set of entities that enables performance monitoring and reporting. Each of these entities also enables network performance assessment against planned goals, performs various aspects of trend analysis, including error rate and cause analysis and Service degradation. Entities in this ABE also enable the traffic trend analysis.	The Service Performance Specification ABE defines measure of the manner in which a Service is functioning.

Service Usage	ServiceUsageSpec	The Service Usage ABE represents the specification of Service Usage types and collects Service Usage records with their characteristics (a.k.a. consumption data).	This is the base entity for defining the Service hierarchy. All Services are characterized as either being related to Products or Resources. This gives rise to the two subclasses of Service: CustomerFacingService (used to realize a Product) and ResourceFacingService (which support the network/infrastructure facing part of the service). Services are defined as being tightly bound to Products. A Product defines the context of the Service, Service and its related entities (e.g., ServiceSpecification, ServiceRole, and so forth) are related to entities in the Resource, Product, and other domains through a set of relationships. A Service represents the object that will be instantiated. Each Service instance can be different; therefore, Service is limited to owning just the attributes, relationships, and constraints that are specific to an instance of a Service. The shared attributes, relationships, and constraints that can be instantiated are defined by a ServiceSpecification. The purpose of this entity is twofold. First, it is used to define attributes, and relationships that are common to all Services. Second, it provides a convenient point to define how Services interact with other parts business entities.

Service Configuration	ServiceConfigSpec	The Service Configuration ABE represents the specification of the different possible Service configurations as well as the Service Configurations instantiated for a Service.	The definition of how a Service operates or functions in terms of CharacteristicSpecification(s) and related ResourceSpec(s) and ServiceSpec(s).

2.3. eTOM L2 - SID ABEs links



2.4. Functional Framework Functions

Function ID	Function Name	Function Description	Aggregate Function Level 1	Aggregate Function Level 2

1135	Technical Solution Policy Design	The Technical Solution Policy Management Function enables to define, and to check consistency of, potentially complex rules to automate the choice of the technical solution (CFS Spec) for a know-how (RFS Spec) or parts of the technical solution. / Criteria can be of various nature and combined: / geographical / depending on the chosen characteristics values of the know-how / depending on other products / know-hows of the customer installed or ordered / depending on the current state of Service Platforms, Network equipment (occupation rate, load balancing, etc.) / depending on Supplier/Partner contracts (volume discounts, etc) / Etc.	Service Specification Development	Technical Solution Policy Management
995	Service Task Item Policy Control Configuration	Define and configure the policies which will be implemented during the Service task item lifecycle.	Service Specification Development	Service Specification Design

1080	Service Specification Change Auditing	Service Specification Change Auditing manages the implications of Service Specifications changes to determine the consequences of any given change. Customer Facing Service Specifications (CFSSpec) changes may impact other CFSSpec according to relationships and also Product Specification that restrict it. Resources Facing Service Specification (RFSSpec) may impact other RFSSpecs according to relationships and / or CFSSpecs it realizes. / The function logs Service Specifications changes and supports the analysis of relationships between Service Specifications. / In addition, it tracks the history of changes in an easy and accessible manner.	Service Specification Development	Service Specification Design

1081	Service Specification Repository Management	Service Specification Repository Management is able to create, modify and delete Service Specification and related entities such as Service Usage Specification. / This includes the ability to manage the state of an entity during its lifecycle (e.g. planned, deployed, in operation, replaced by, locked). / It includes Service Specifications retrieval, integrity rules check and versioning management. /	Service Specification Development	Service Specification Design
1084	Know-How Specification Design	Know-How Specification Design provides the means to describe the Customer Facing Service Specifications including constraints, characteristics and types of usages. / It also identifies Technical Solutions usable for the Know-How and rules to find the Technical Solution.	Service Specification Development	Service Specification Design

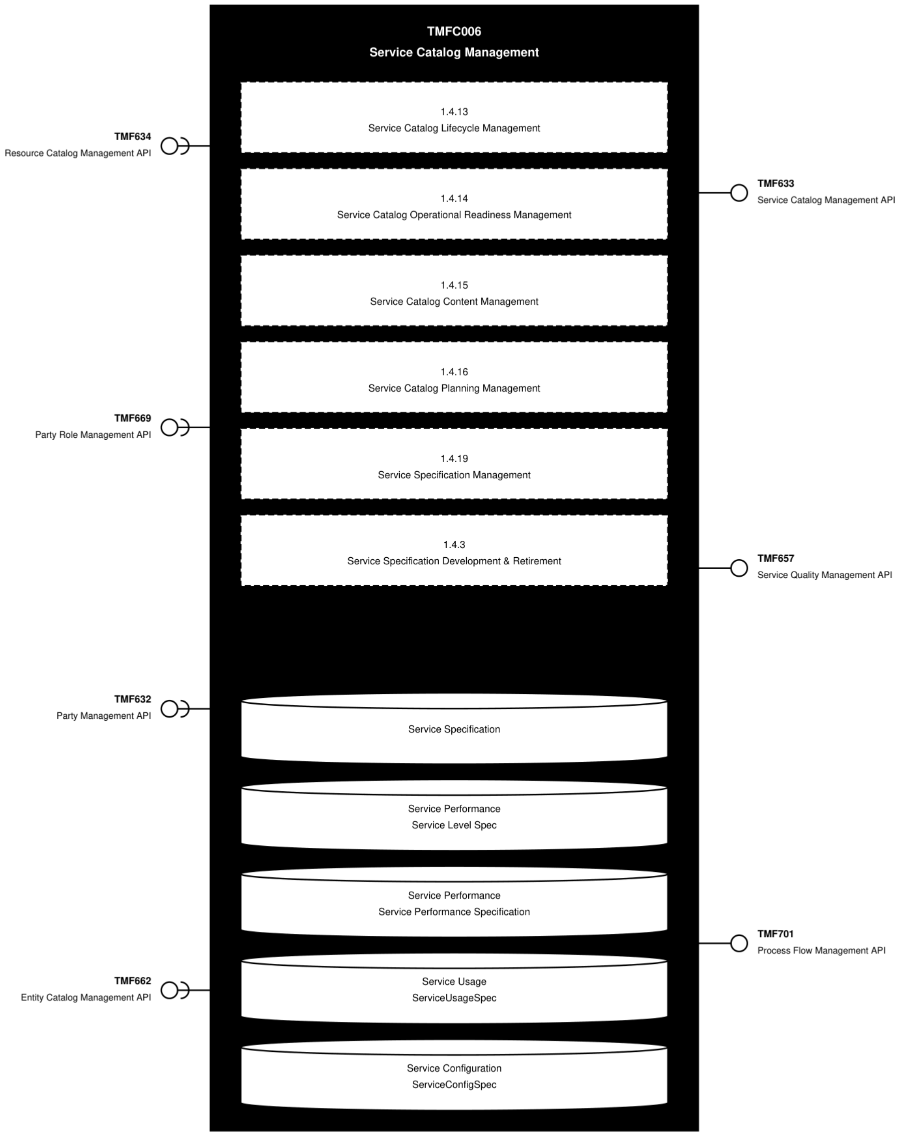
1085	Technical Solution Design	Technical Solution Design provides the means to describe Resource Facing Service Specifications (a.k.a. RFSSpec) including constraints, characteristics and types of usages. / It also identifies Resource Specifications used for each Technical Solution Specification (a.k.a. RFSSpec).	Service Specification Development	Service Specification Design
1086	Service Specification to Supplier Product Specification Relationship Design	Service Specification to Supplier Product Specification Relationship Design identifies, when know-how, technical solutions or part of it are not realized by the CSP, the Supplier Product Specification used to implement and corresponding rules.	Service Specification Development	Service Specification Design
897	Building Access Control	Building Access Control checks, stops or allow physical access to facilities according to access roles and rules.	Identification and Permission Management	Permission Control

900	Authorization Control Management	Authorization Control Function controls permissions according to roles and related rules. / It consists in evaluating if a requester is granted the permission to act by providing the required evidence. The evidence corresponds to the condition specified for each right (for instance keying the correct password to use a specific mailbox). If the action is protected via a right which is assigned (possibly via a role) to a person then the person has to be identified to retrieve their rights and verify if the request to act can be granted.	Identification and Permission Management	Permission Control

3. TM Forum Open APIs & Events

The following part covers the APIs and Events; this part is split in 3: - List of Exposed APIs - This is the list of APIs available from this component. - List of Dependent APIs - In order to satisfy the provided API, the component could require the usage of this set of required APIs. - List of Events (generated & consumed) - The events which the component may generate are listed in this section along with a list of the events which it may consume. Since there is a possibility of multiple sources and receivers for each defined event.

3.1. API Context Diagram

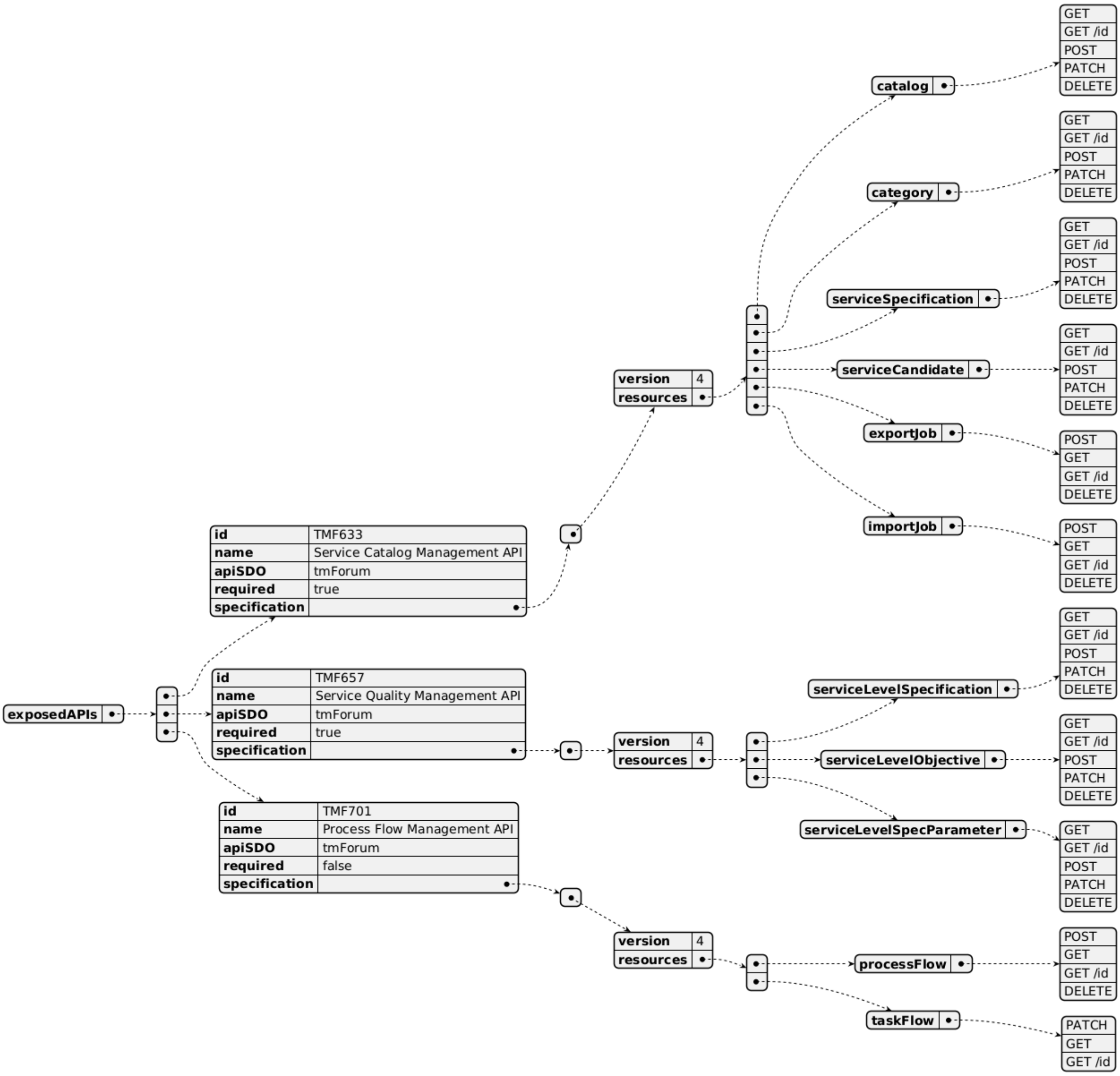


(SVG source: [TMFC006_API_Context.svg]() — hand-drawn rather than PlantUML for this one diagram, since PlantUML's automatic layout couldn't give each API its own straight, individually-anchored connector at a chosen height. Generated by the `component-specification-documentation` *skill's* `scripts/render_api_context_svg.py`*.)*

3.2. Exposed APIs

API ID	API Name	Mandatory / Optional	API Version	Resource	Operations

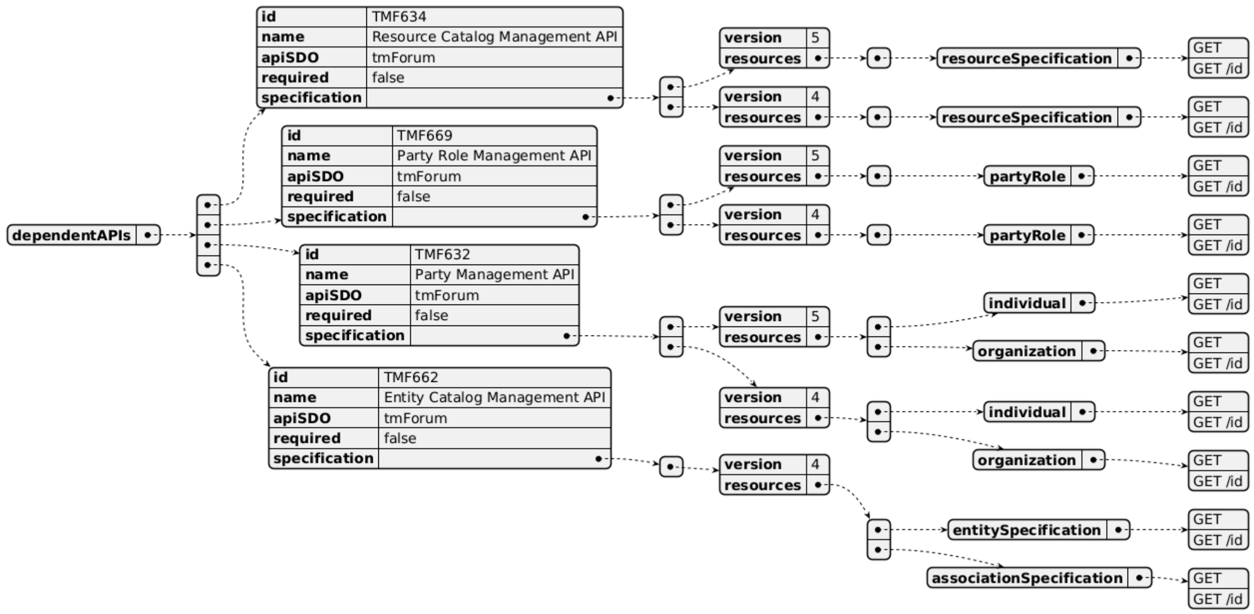
TMF633	Service Catalog Management API	Mandatory	4	catalog category serviceSpecification serviceCandidate exportJob importJob	GET GET /id POST PATCH DELETE
TMF657	Service Quality Management API	Mandatory	4	serviceLevelSpecification serviceLevelObjective serviceLevelSpecParameter	GET GET /id POST PATCH DELETE
TMF701	Process Flow Management API	Optional	4	processFlow taskFlow	POST GET GET /id DELETE PATCH



3.3. Dependent APIs

API ID	API Name	API Version	Mandatory / Optional	Resource	Operations
TMF634	Resource Catalog Management API	5	Optional	resourceSpecification	GET GET /id
TMF634	Resource Catalog Management API	4	Optional	resourceSpecification	GET GET /id
TMF669	Party Role Management API	5	Optional	partyRole	GET GET /id

TMF669	Party Role Management API	4	Optional	partyRole	GET GET /id
TMF632	Party Management API	5	Optional	individual organization	GET GET /id
TMF632	Party Management API	4	Optional	individual organization	GET GET /id
TMF662	Entity Catalog Management API	4	Optional	entitySpecification associationSpecification	GET GET /id



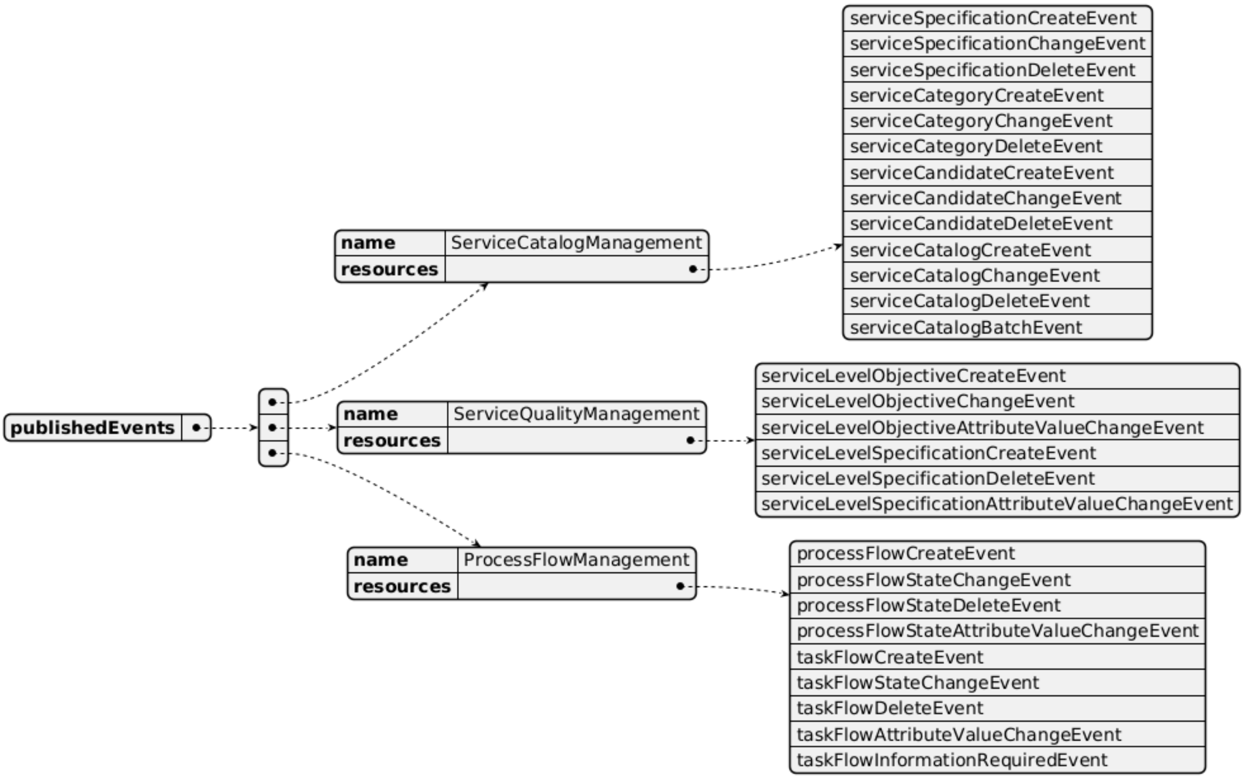
3.4. Events

The diagram illustrates the Events which the component may publish and the Events that the component may subscribe to and then may receive. Both lists are derived from the APIs listed in the preceding sections.

Published Events

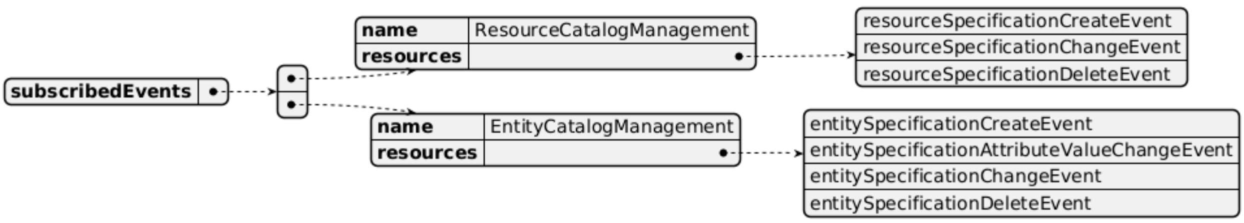
API ID	API Name	Event Resources

TMF633	Service Catalog Management API	serviceSpecificationCreateEvent serviceSpecificationChangeEvent serviceSpecificationDeleteEvent serviceCategoryCreateEvent serviceCategoryChangeEvent serviceCategoryDeleteEvent serviceCandidateCreateEvent serviceCandidateChangeEvent serviceCandidateDeleteEvent serviceCatalogCreateEvent serviceCatalogChangeEvent serviceCatalogDeleteEvent serviceCatalogBatchEvent
TMF657	Service Quality Management API	serviceLevelObjectiveCreateEvent serviceLevelObjectiveChangeEvent serviceLevelObjectiveAttributeValueChangeEvent serviceLevelSpecificationCreateEvent serviceLevelSpecificationDeleteEvent serviceLevelSpecificationAttributeValueChangeEvent
TMF701	Process Flow Management API	processFlowCreateEvent processFlowStateChangeEvent processFlowStateDeleteEvent processFlowStateAttributeValueChangeEvent taskFlowCreateEvent taskFlowStateChangeEvent taskFlowDeleteEvent taskFlowAttributeValueChangeEvent taskFlowInformationRequiredEvent



Subscribed Events

API ID	API Name	Event Resources
TMF634	Resource Catalog Management API	resourceSpecificationCreateEvent resourceSpecificationChangeEvent resourceSpecificationDeleteEvent
TMF662	Entity Catalog Management API	entitySpecificationCreateEvent entitySpecificationAttributeValueChangeEvent entitySpecificationChangeEvent entitySpecificationDeleteEvent



4. Machine Readable Component Specification

Refer to the ODA Component Directory on the TM Forum website for the machine-readable component specification files for this component.

5. References

5.1. TMF Standards related versions

Standard	Version(s)
eTOM	v23.0
SID	v25.0
Functional Framework	v23.0

5.2. Jira References

No open Jira issues currently listed for this component.

5.3. Further resources

1. IG1228: please refer to IG1228 for defined use cases with ODA components interactions.
2. TMF633 Service Catalog Management API User Guide v4.0.0

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date	Modified by	Description of changes
1.0.0	29-Mar-2022	Kamal Maghsoudlou, Gaetano Biancardi, Sylvie Demarest	Final edits prior to publication

1.0.1	25-Jul-2023	Ian Turkington	No content changed, simply a layout change to match template 3. Separated the YAML files to a managed repository.
1.0.1	14-Aug-2023	Amaia White	Final edits prior to publication
1.1.0	03-May-2024	Amaia White	Final edits prior to publication
1.2.0	12-Nov-2024	Gaetano Biancardi	TMF688 removed from the core specification, moved to supporting functions; TMF672 removed from the core specification, moved to supporting functions; API version, only major version to be specified
1.2.0	26-Nov-2024	Amaia White	Final edits prior to publication
1.2.1	08-Jul-2025	Rosie Wilson	Updates to description per agreed YAML updated on 03-Jun-2025; updated exposed API TMF657 to mandatory in the table to align with the diagram and YAML
1.2.1	15-Jul-2025	Rosie Wilson	Final edits prior to publication

6.1.2. Release History

Release Status	Date Modified	Modified by	Description of changes
Pre-production	29-Mar-2022	Goutham Babu	Initial release
Production	20-May-2022	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14-Aug-2023	Amaia White	New release 1.0.1
Production	06-Oct-2023	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	02-May-2024	Hugo Vaughan	Version 1.1.0 alignment to Frameworks 23.5; Service Test descoped to enable the definition of an additional Component
Pre-production	02-May-2024	Amaia White	New release 1.1.0

Production	28-Jun-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	26-Nov-2024	Amaia White	New release 1.2.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	15-Jul-2025	Rosie Wilson	New release 1.2.1

6.2. Acknowledgements

Team Member	Company	Role
Elisabeth Andersson	MATRIXX	Editor
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Gaetano Biancardi	Accenture	Additional Input
Sylvie Demarest	Orange	Additional Input
Ian Turkington	TM Forum	Additional Input
Abinash Vishwakarma	Netcracker	Reviewer
Jose Macaluso	Xacria	Reviewer
Kamal Maghsoudlou	Ericsson	Reviewer
Milind Bhagwat	BT	Reviewer
Ritu Arora	BT	Reviewer
Matteo Destino	Accenture	Reviewer